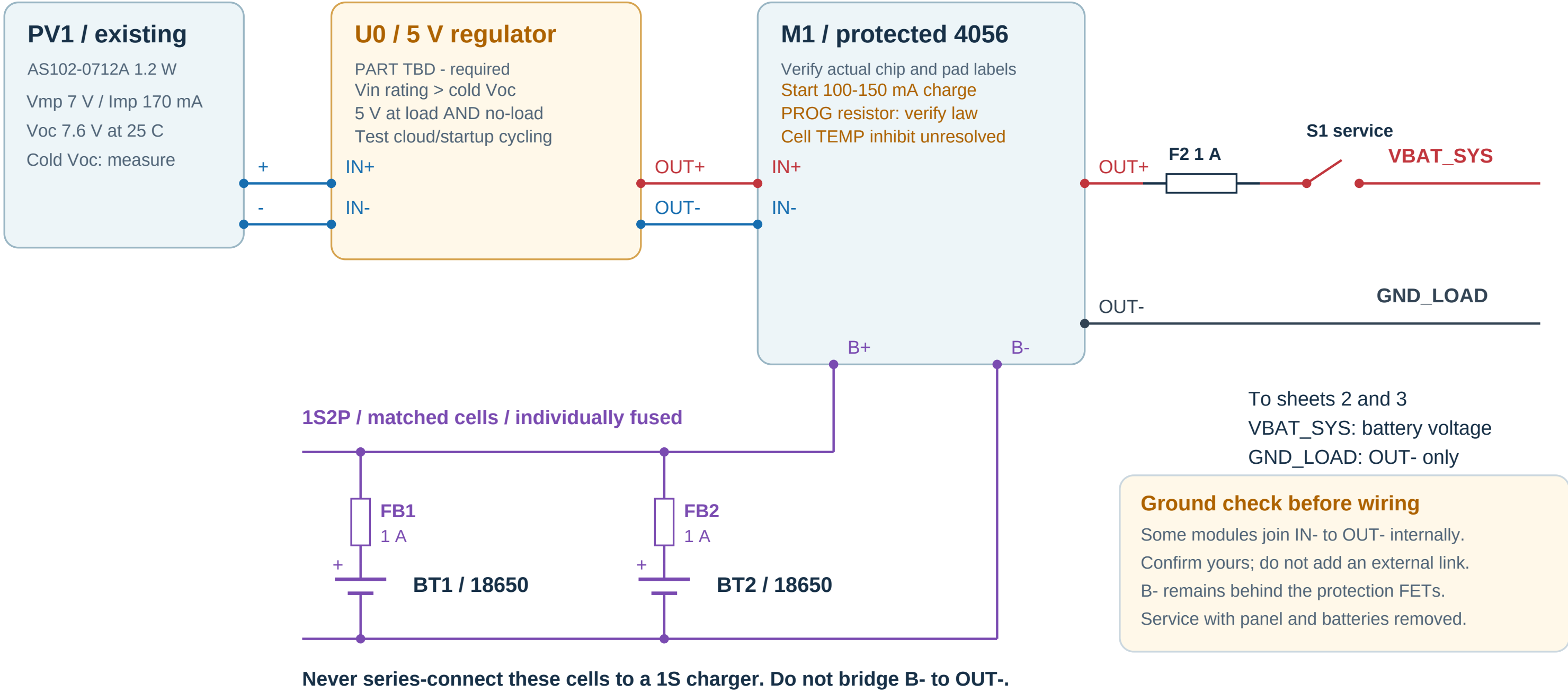


# Power, charging & battery

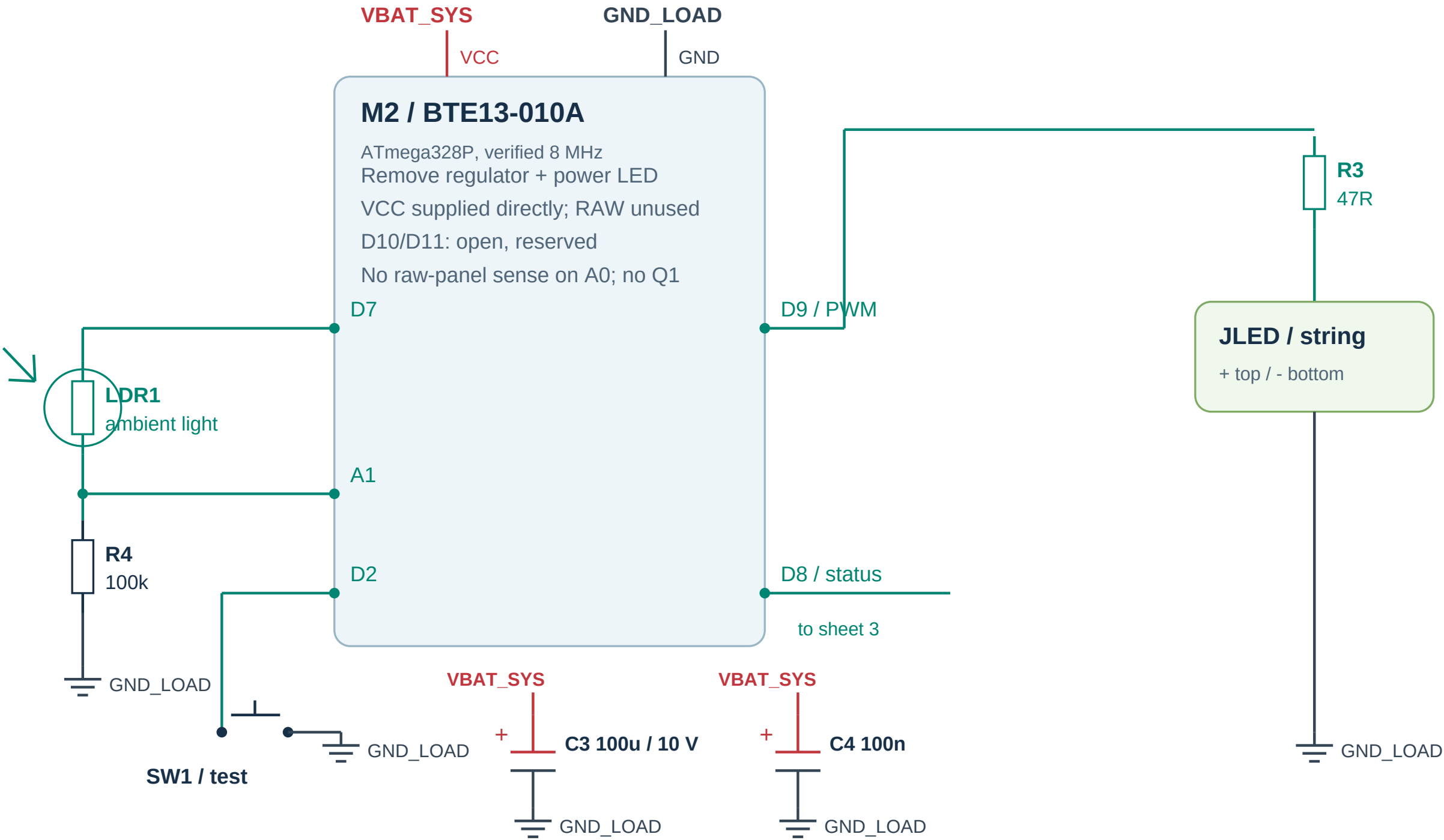
Terminal-level schematic. Amber blocks need part selection and bench qualification. All loads connect to protected outputs.



No pair of series diodes is accepted as overvoltage protection. Unattended charging awaits cell-temperature protection.

# Light controller & LED driver

Reused 16 MHz Pro Mini converted to internal 8 MHz. Switched LDR; dusk to dawn at 100% command, with R3 setting the current.



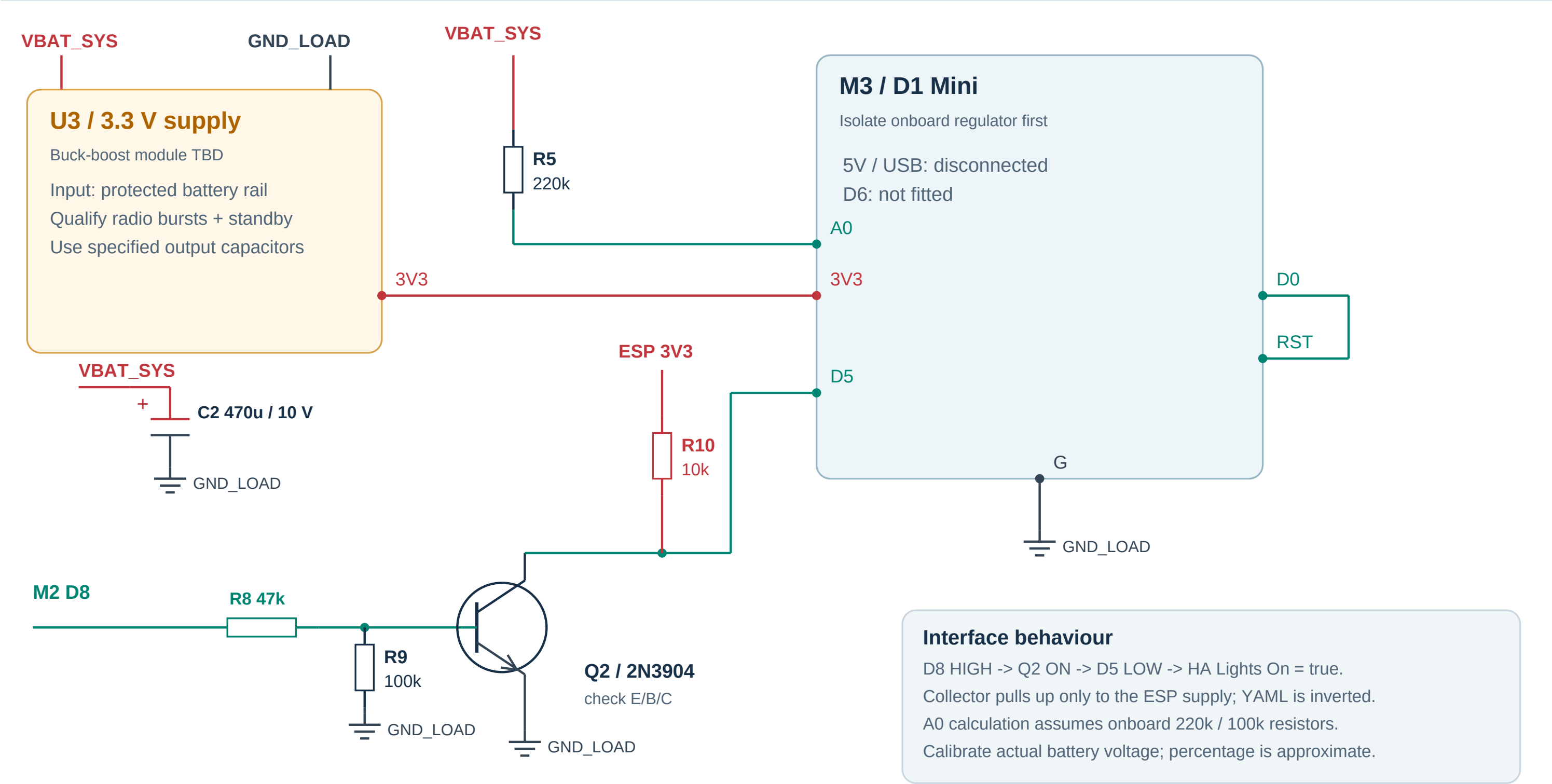
**No Q1, R6, R7 or F1**

Measured string: 2.56 V drop  
 $I = (VBAT - 2.56) / (R3 + 30R)$   
21 / 15 / 11 mA at  
4.20 / 3.70 / 3.40 V  
Fit 56R to stay under 20 mA  
Pin max 40 mA absolute  
Fit R3 at the board end

All ground symbols on this sheet mean OUT-. R3 sets the peak current; duty changes the average only. Verify current at 4.20 V and 100% duty.

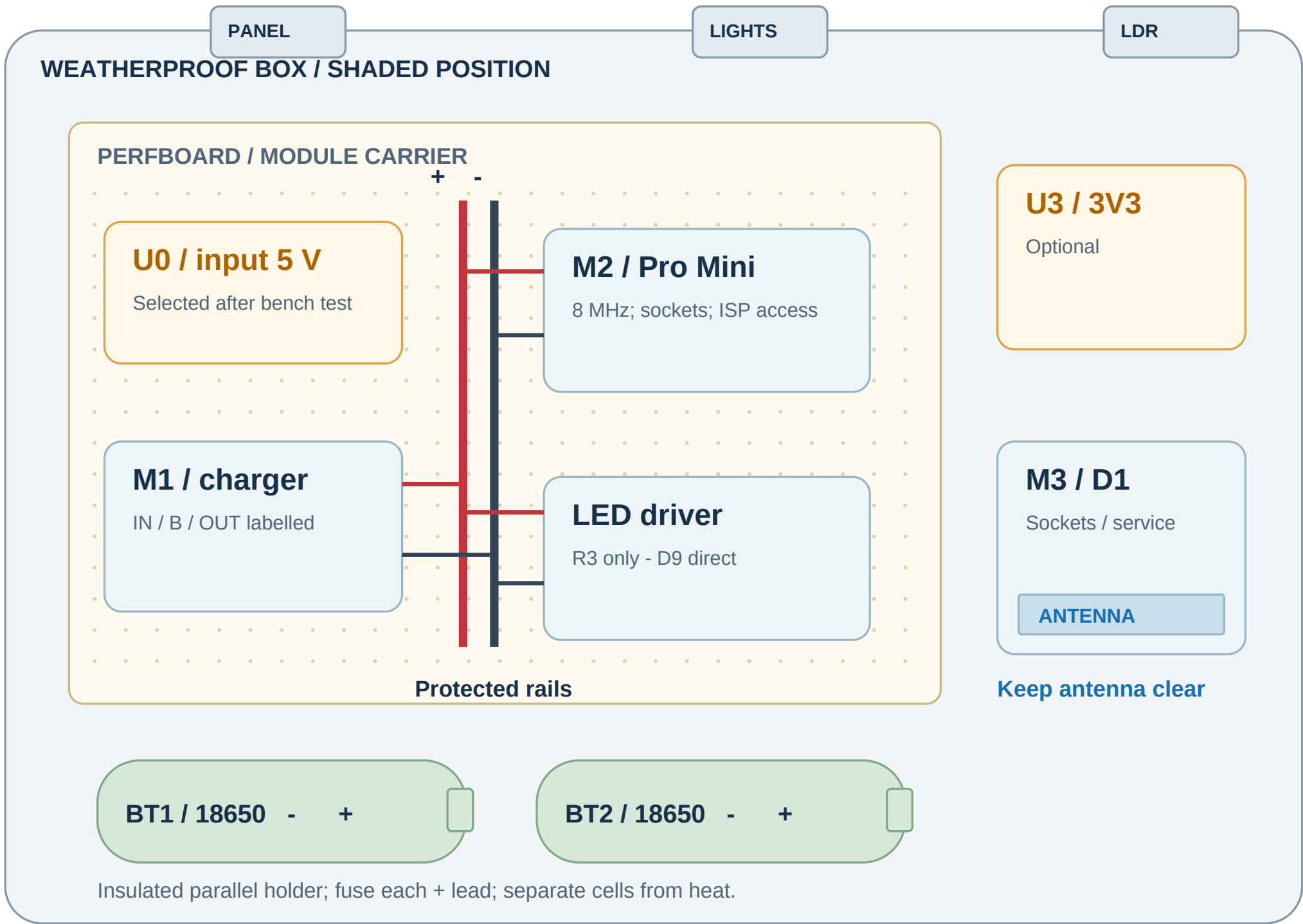
# Low-power telemetry reference (D1 Mini)

Retained for the later production profile. This is not the always-on D1 Mini UART test system on sheet 6.



# What to assemble inside the enclosure

Top-down placement concept, not a drilling template. Use terminal labels and the wire schedule for electrical connections.



## Build in this order

- 1 Qualify panel regulator + charger.
- 2 Convert and test the Pro Mini.
- 3 Add R3 (47R) and light string - no MOSFET.
- 4 Add LDR; check dusk/dawn + LVC.
- 5 Assemble the protected, fused pack.
- 6 Add optional telemetry and measure.

## Label every connector

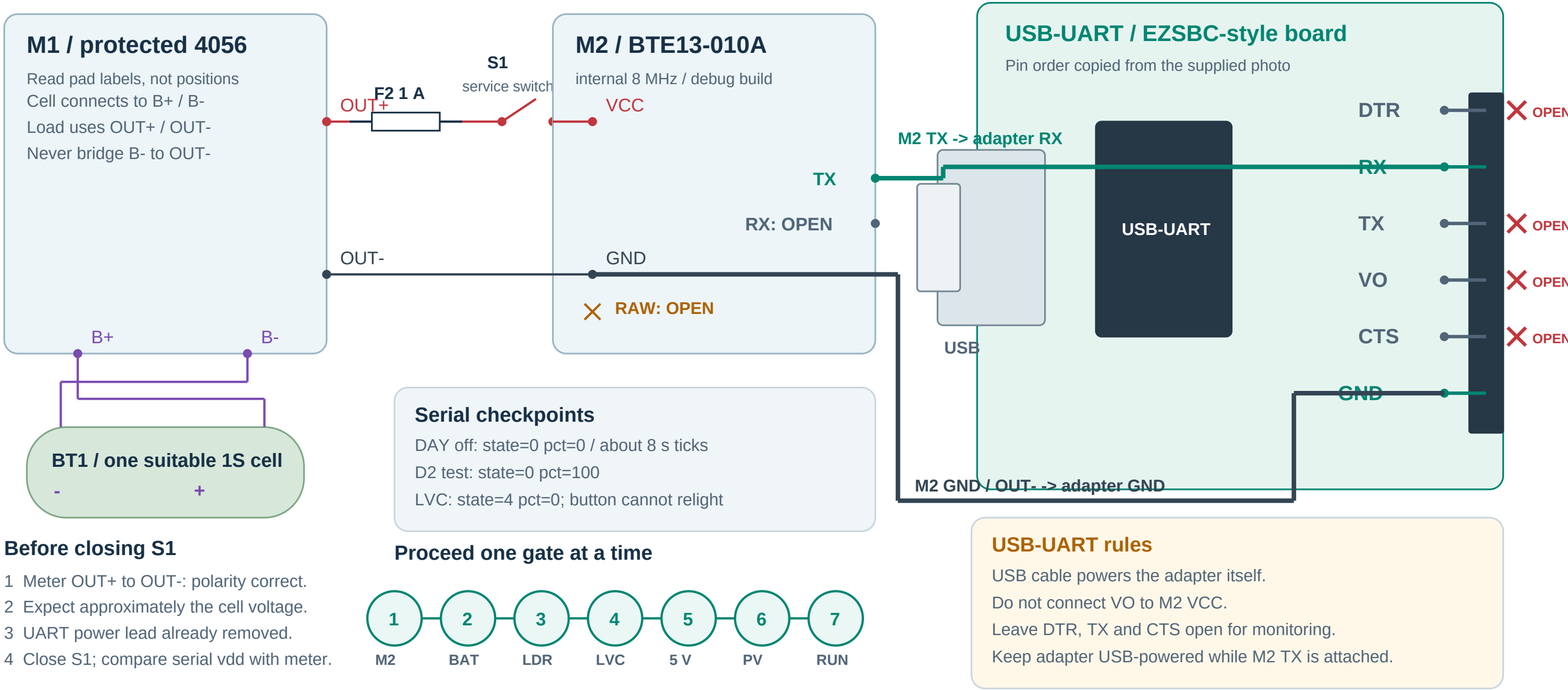
RED: protected positive / VBAT\_SYS  
BLACK: load return / OUT-  
BLUE: panel input pair  
PURPLE: battery-only connections  
GREEN: logic / sensor signals

## Before closing the lid

Strain relief, polarity, cell insulation.  
No exposed underside copper.  
Charger/cell temperature tests passed.  
Log charge, load and actual brightness.

# Staged power-up & serial diagnostics

Battery powers the controller. The USB-UART listens only: connect RX and GND; leave its power and control pins open.



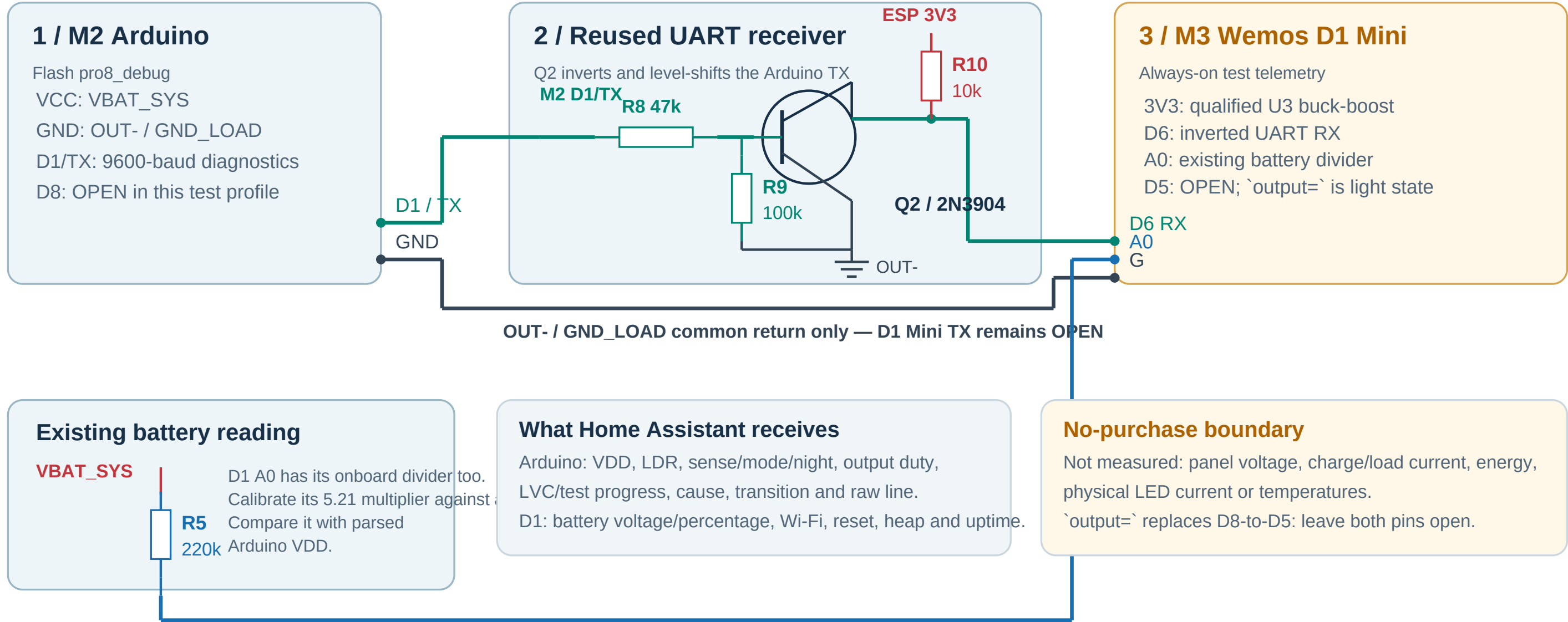
**SERIAL IS NOT A CHARGE METER: verify panel input, regulated 5 V, battery voltage and charge current with instruments.**

# Development telemetry (Wemos D1 Mini)

TEST SYSTEM ONLY: always-on Wi-Fi and UART capture. Reuses the existing status transistor parts; no INA226 or new telemetry module. Production D1 profile remains on sheet 3.

**Read this first**

Always-on Home Assistant test configuration: Q2/R8/R9/R10 become a safe, inverted UART receiver. The D1 Mini deep-sleep production reference is sheet 3.



Test rules: run pro8\_debug • no USB-UART adapter on M2 • keep D1 awake • use solar-lights-lite.yaml later for production

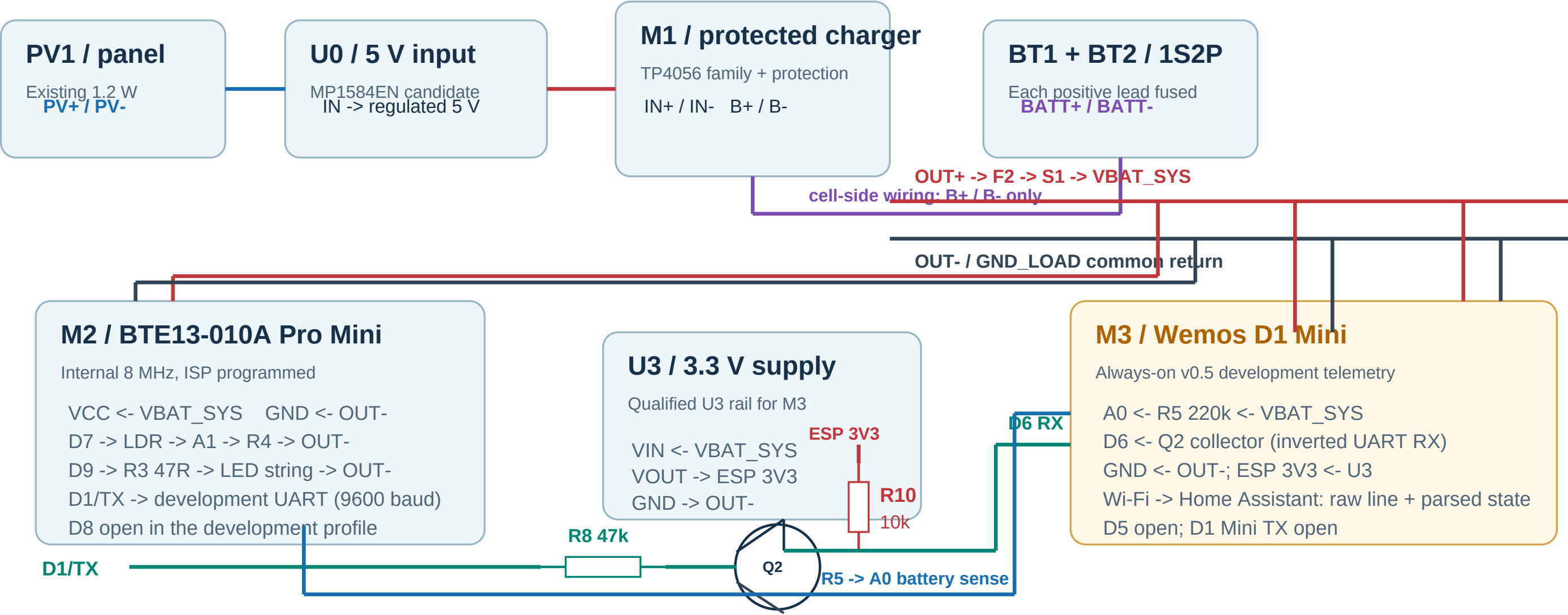


# Current architecture schematic

Live v0.5 bench wiring. The development receiver captures the Arduino UART in Home Assistant; it is not a current or energy meter.

### How to use this sheet

Read this as the system-level schematic. Follow sheets 1, 2 and 6 for terminal-level wiring. The editable KiCad source in kicad\_current/ uses this same D1/TX-to-D6 development route.



### Current visible in Home Assistant

Arduino: state, VDD, LDR, output, low-voltage progress and raw diagnostics.  
D1: battery, Wi-Fi, reset reason, heap and uptime.

### Still measured with instruments

Panel/charge voltage and charge/load current or energy.  
Physical LED current and temperature.